

*ident*, *x* (\* identifiers \*)

|                       |     |                                                          |                                    |
|-----------------------|-----|----------------------------------------------------------|------------------------------------|
| <i>exp</i> , <i>e</i> | ::= |                                                          | expressions like <i>exp</i>        |
|                       |     | <i>x</i>                                                 | variable named <i>x</i>            |
|                       |     | ()                                                       | unit value                         |
|                       |     | ( <i>exp</i> , <i>exp'</i> )                             | pair of <i>exp</i> and <i>exp'</i> |
|                       |     | <b>function</b> <i>x</i> : type $\rightarrow$ <i>exp</i> | bind <i>x</i> in <i>exp</i>        |
|                       |     | <i>exp exp'</i>                                          |                                    |
|                       |     | { <i>exp</i> / <i>x</i> } <i>exp'</i>                    | M                                  |
|                       |     | ( <i>exp</i> )                                           | M                                  |

|                |     |                          |
|----------------|-----|--------------------------|
| type, <i>t</i> | ::= |                          |
|                |     | unit                     |
|                |     | type $\star$ type'       |
|                |     | type $\rightarrow$ type' |

|          |     |                                            |
|----------|-----|--------------------------------------------|
| <i>E</i> | ::= |                                            |
|          |     | <b>empty</b>                               |
|          |     | <i>E</i> , <i>x</i> : <i>t</i>             |
|          |     | <i>E</i> , <i>x</i> : <i>t</i> , <i>E'</i> |

|                       |     |                                                              |
|-----------------------|-----|--------------------------------------------------------------|
| <i>val</i> , <i>v</i> | ::= |                                                              |
|                       |     | <i>x</i>                                                     |
|                       |     | ()                                                           |
|                       |     | ( <i>val</i> , <i>val'</i> )                                 |
|                       |     | <b>function</b> <i>x</i> : <i>t</i> $\rightarrow$ <i>exp</i> |

|                |     |                                |
|----------------|-----|--------------------------------|
| <i>formula</i> | ::= |                                |
|                |     | <i>judgement</i>               |
|                |     | $\neg$ ( <i>formula</i> )      |
|                |     | <i>x</i> $\in$ dom( <i>E</i> ) |
|                |     | M                              |

|                  |     |                   |
|------------------|-----|-------------------|
| <i>terminals</i> | ::= |                   |
|                  |     | $\star$           |
|                  |     | $\rightarrow$     |
|                  |     | <b>function</b>   |
|                  |     | -                 |
|                  |     | $\vdash$          |
|                  |     | $\longrightarrow$ |
|                  |     | {                 |
|                  |     | }                 |

|              |     |                                       |                                                              |
|--------------|-----|---------------------------------------|--------------------------------------------------------------|
| <i>Jtype</i> | ::= |                                       |                                                              |
|              |     | <i>E</i> $\vdash$ <b>ok</b>           | M                                                            |
|              |     | <i>E</i> $\vdash$ <i>x</i> : <i>t</i> | M                                                            |
|              |     | <i>E</i> $\vdash$ <i>e</i> : <i>t</i> |                                                              |
|              |     |                                       | term <i>e</i> has type <i>t</i> in type environment <i>E</i> |

|            |     |                                      |
|------------|-----|--------------------------------------|
| <i>Jop</i> | ::= |                                      |
|            |     | <i>e</i> $\longrightarrow$ <i>e'</i> |

|                  |     |  |
|------------------|-----|--|
| <i>judgement</i> | ::= |  |
|------------------|-----|--|

|                |       |             |
|----------------|-------|-------------|
|                |       | $Jtype$     |
|                |       | $Jop$       |
| $user\_syntax$ | $::=$ |             |
|                |       | $ident$     |
|                |       | $exp$       |
|                |       | <b>type</b> |
|                |       | $E$         |
|                |       | $val$       |
|                |       | $formula$   |
|                |       | $terminals$ |

$E \vdash \mathbf{ok}$

$$\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \quad \text{EOK\_EMPTY}$$

$$\frac{E \vdash \mathbf{ok} \quad \neg(x \in \text{dom}(E))}{E, x : \mathbf{t} \vdash \mathbf{ok}} \quad \text{EOK\_IDENT}$$

$E \vdash x : \mathbf{t}$

$$\frac{E, x : \mathbf{t}, E' \vdash \mathbf{ok}}{E, x : \mathbf{t}, E' \vdash x : \mathbf{t}} \quad \text{EXT\_IDENT}$$

$E \vdash e : \mathbf{t}$     term  $e$  has type  $\mathbf{t}$  in type environment  $E$

$$\frac{E \vdash x : \mathbf{t}}{E \vdash x : \mathbf{t}} \quad \text{EET\_IDENT}$$

$$\frac{E \vdash \mathbf{ok}}{E \vdash () : \mathbf{unit}} \quad \text{EET\_UNIT}$$

$$\frac{E \vdash e_1 : \mathbf{t}_1 \quad E \vdash e_2 : \mathbf{t}_2}{E \vdash (e_1, e_2) : \mathbf{t}_1 \star \mathbf{t}_2} \quad \text{EET\_PAIRPAIRING}$$

$$\frac{E, x : \mathbf{t}_1 \vdash e : \mathbf{t}_2}{E \vdash \mathbf{function} \ x : \mathbf{t}_1 \rightarrow e : \mathbf{t}_1 \rightarrow \mathbf{t}_2} \quad \text{EET\_FUN}$$

$$\frac{E \vdash e : \mathbf{t}_1 \rightarrow \mathbf{t}_2 \quad E \vdash e' : \mathbf{t}_1}{E \vdash e \ e' : \mathbf{t}_2} \quad \text{EET\_APP}$$

$e \longrightarrow e'$

$$\frac{}{(\mathbf{function} \ x : \mathbf{t} \rightarrow e) \ v \longrightarrow \{v/x\}e} \quad \text{JO\_RED\_FUN}$$

$$\frac{e_1 \longrightarrow e'_1}{e_1 \ e_2 \longrightarrow e'_1 \ e_2} \quad \text{JO\_RED\_CONTEXT1}$$

$$\frac{e_2 \longrightarrow e'_2}{v_1 \ e_2 \longrightarrow v_1 \ e'_2} \quad \text{JO\_RED\_CONTEXT2}$$

$$\frac{e_1 \longrightarrow e'_1}{(e_1, e_2) \longrightarrow (e'_1, e_2)} \quad \text{JO\_RED\_CONTEXT3}$$

$$\frac{e_2 \longrightarrow e'_2}{(v_1, e_2) \longrightarrow (v_1, e'_2)} \quad \text{JO\_RED\_CONTEXT4}$$

Definition rules: 13 good 0 bad  
Definition rule clauses: 27 good 0 bad